

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

Nov 2012

CL201.01 Surveying

Date: 27.11.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What are the errors that may occur in plane tabling? [04]
(b) Explain any one procedure to measure the depth of water. [03]
- Q - 2 (a) Explain orientation by back sighting method. [04]
(b) Explain procedure to measure Magnetic bearing with the help of theodolite. [04]
(c) Calculate and correct the latitudes and departures for the following data [06]

Line	Length	RB
AB	105.8	N 40° 45' W
BC	142.5	N 51° 30' E
CD	188.8	S 48° 15' E
DA	188.9	S 76° 45' W

OR

- Q - 2 (a) Write advantages and disadvantages of plane table survey. [04]
(b) Explain procedure to measure deflection angle. [04]
(c) Calculate length of RS and ST from the following data. [06]

Line	Length	RB
PQ	178.6	S 52° 30' E
QR	228.4	N 48° 45' E
RS	?	N 18° 15' W
ST	?	S 78° 30' W
TP	238.8	S 32° 30' W

- Q - 3 (a) What is the principal of plane tabling? Explain procedure of centering and leveling in plane tabling. [04]
(b) Write the function of the following [04]
i) Foot Screw ii) Tangent screw iii) Graphic display iv) ATR
(c) Write advance functions in total station. Explain any two in detail. [06]

OR

- Q - 3 (a) List the methods of plane tabling explain the traversing method in detail [04]
 (b) Define following [04]
 i) Telescope normal ii) Line of collimation iii) Total station iv) EDM
 (c) Explain different types of EDM and basic difference between them. [06]

SECTION – II

- Q - 4 (a) Explain procedure of profile leveling and plotting of longitudinal section. [04]
 (b) Define [03]
 i) Contour line ii) Contour interval iii) Horizontal equivalent
 Q - 5 (a) Two tangents intersect at chainage 1450 m. The angle of intersection is 140° . [08]
 Calculate all data necessary for setting out a curve of radius of 300 m by the deflection angle method. The peg intervals may be taken as 20m.
 (b) Calculate area by trapezoidal rule and Simpson's rule for the following data [06]

Chainage(m)	0	5	10	15	20	25	35	45	55	65
Offset (m)	2.5	3.8	8.4	7.6	10.5	9.3	5.8	7.8	6.9	8.4

OR

- Q - 5 (a) A compound curve is made up of two arcs of radii 380m and 520m. The deflection [08]
 angle of the combined curve is 105° and that of first arc of radius 380m is 58° . The chainage of the first tangent point is 848.55m. Find the chainage of the point of intersection, common tangent point, and forward tangent point.
 (b) Explain components of digital planimeter. [06]
 Q - 6 (a) What is setting out of building? Explain procedure to set out building. [04]
 (b) The Ground levels at 20-m intervals on a stretch of ground are as given below. The [06]
 formation level at zero chainage is 212.66 m. and the formation goes at a longitudinal upward slope of 1:40. Find the volume of the earthwork in this stretch, if formation width is 6m and the side slope is 2:1. The ground is level in the transverse direction. Use trapezoidal and Prismoidal rule.

Chainage(m)	0	20	40	60	80	100	120
G.L(m)	214.2	214.8	215.1	216.1	216.9	217.4	218.2

- (c) List the various methods of setting out a simple circular curve. Explain any one [04]
 method in detail.

OR

- Q - 6 (a) Explain procedure to set out of pipelines and sewers. [04]
 (b) A road embankment is 8 m wide and 200m in length, at the formation level, with a [06]

side slope of 1.5:1. The embankment has a rising gradient of 1 in 200m. The ground levels at every 100 m along the centre line are as follows. The formation level of zero chainage is 166m. Calculate the volume of earth work. By trapezoidal rule and Prismoidal rule

Chainage(m)	0	100	200	300	400
R.L(m)	164.5	165.2	166.8	167	167.2

- (c) Classify the different types of curves with neat sketches.

[04]
